Chapter Notes: Changes Around Us - Physical and Chemical

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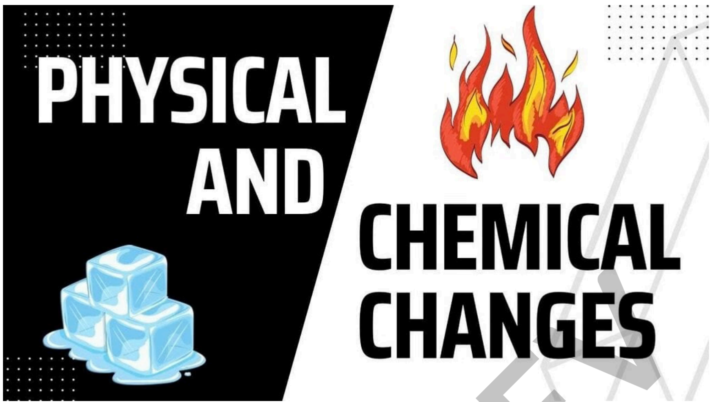
Introduction

We observe many changes around us every day. Ice melts into water, flowers open from buds, fruits change colour and smell, and cold water becomes warm over time. These changes alter how things look, smell, feel, sound or taste. By using our five senses - **sight**, **smell**, **touch**, **hearing** and **taste** - we notice and describe these changes.

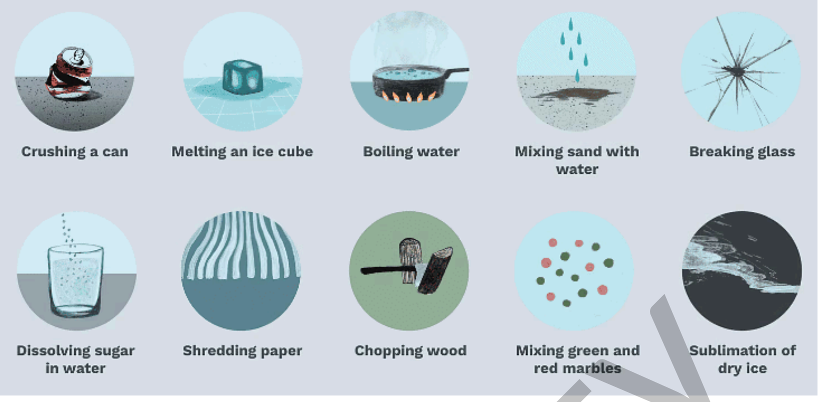


In this chapter we learn to distinguish between two broad kinds of changes: **physical changes** and **chemical changes**. We study everyday examples, simple tests to detect some chemical changes and why some changes are reversible while others are not.

Physical Change Vs Chemical Change

Physical Change

Definition: A physical change is one in which the appearance, shape, size or state of a substance changes but its chemical composition stays the same. No new substance is formed.



Physical Changes

Key features of physical changes

- Only physical properties such as shape, size or state change.
- The chemical composition of the substance remains unchanged.
- Physical changes are often reversible by simple physical means, though not always.

Examples of physical changes

- **Paper folding:** Folding paper changes its form; unfolding restores the sheet.

- **Balloon inflation:** Inflating a balloon stretches it; letting the air out returns it toward the original shape. If a balloon bursts, the rubber is still the same material but the original shape cannot be restored.

Crushing chalk: Crushing chalk into powder changes its form but not its chemical identity; the original piece usually cannot be reformed by simple methods.

Change of states of water: Ice → water → steam are physical changes because the substance remains **water**.

MULTIPLE CHOICE QUESTION

Try yourself: What is a physical change?

A A change that creates a new substance.

B A change in appearance without new material.

C A change that cannot be undone.

D A change that occurs only in nature.

View Solution

Chemical Change

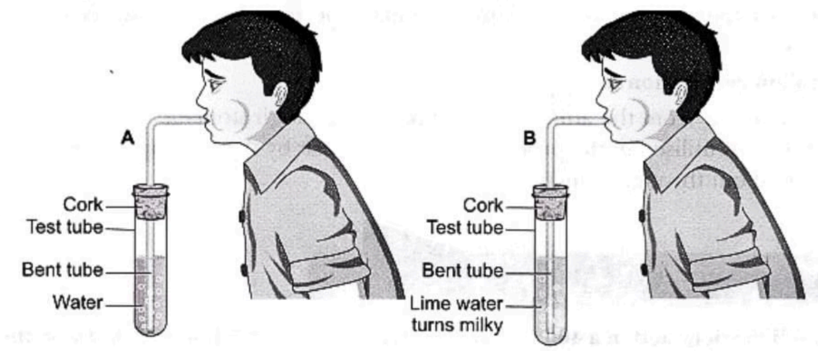
Definition: A chemical change (or chemical reaction) is a process in which one or more new substances with different properties are formed. Chemical changes are usually not easily reversed by simple physical methods.

Chemical Changes

Examples and explanations

1. Blowing air into lime water

When air from our breath containing **carbon dioxide** is bubbled through **lime water** (a solution of calcium hydroxide), a chemical reaction occurs and a new substance forms.



The lime water turns milky because a white insoluble solid, **calcium carbonate**, is formed. This white material later settles as a precipitate. Water is also produced.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

This experiment shows a new substance is produced, so the process is a chemical change.

2. Vinegar and baking soda

When **vinegar** (acetic acid) is mixed with **baking soda** (sodium bicarbonate), a vigorous chemical reaction produces bubbling or fizzing due to the evolution of **carbon dioxide gas**.

The gas produced will turn lime water milky if bubbled through it, confirming that the gas is the same **carbon dioxide** detected in many chemical and biological processes.

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + CO_2 + H_2O$$

The bubbling is direct evidence of gas formation; new chemical species (for example, sodium acetate) are also produced.

MULTIPLE CHOICE QUESTION

Try yourself: What happens to lime water when carbon dioxide is blown into it?

A It evaporates.

B It turns milky.

C It remains unchanged.

D It becomes fizzy.

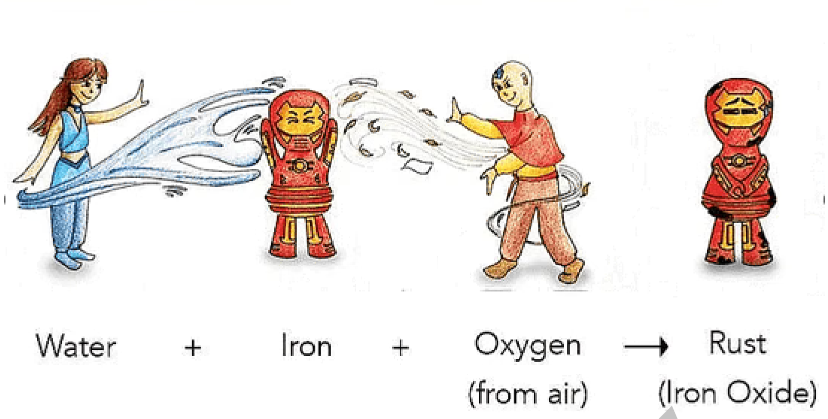
View Solution

Some Other Processes Involving Chemical Changes

Rusting

Rusting is a chemical change in which iron reacts with oxygen and moisture to form **iron oxide** (rust), a new substance that is brown and flaky.

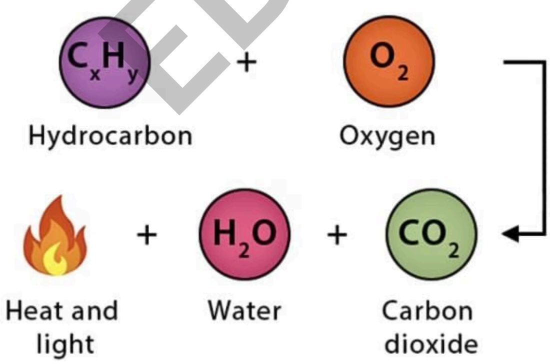
Rust weakens iron objects such as nails, tools and bridges because the new substance has different mechanical and chemical properties from iron.



Rusting of Iron

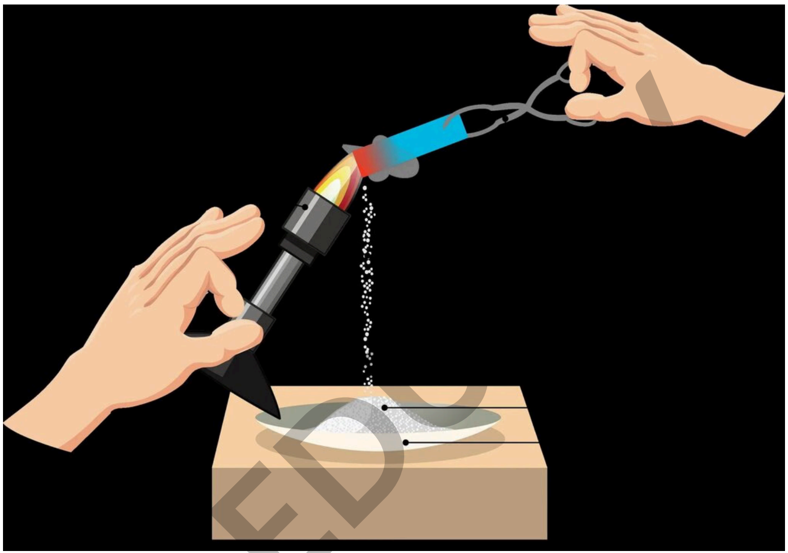
Combustion

Combustion is a chemical reaction in which a substance reacts with oxygen to produce heat and often light. Materials that burn are called **combustible substances** (examples: wood, paper, cotton, kerosene).



For example, burning a magnesium ribbon in air produces a white solid, **magnesium oxide**, showing that new substances are produced during combustion.

Combustion reactions release energy as heat and sometimes light.



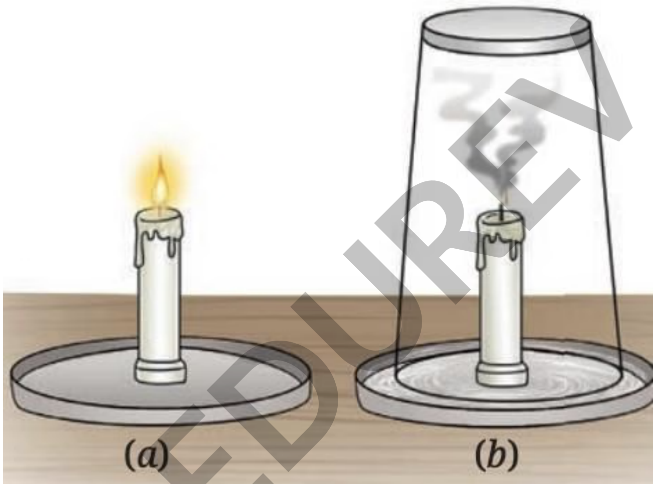
The chemical reaction for burning magnesium (balanced) is shown below.

$$2Mg + O_2 \rightarrow 2MgO$$

Oxygen's role in combustion

Oxygen from the air is essential for most combustion reactions.
If a burning candle or other fire is covered so it cannot get fresh air, the flame goes out because oxygen is used up and cannot be replenished.

During combustion, carbon in the fuel reacts with oxygen to form **carbon dioxide**; this gas can be detected by passing it through lime water, which turns milky.

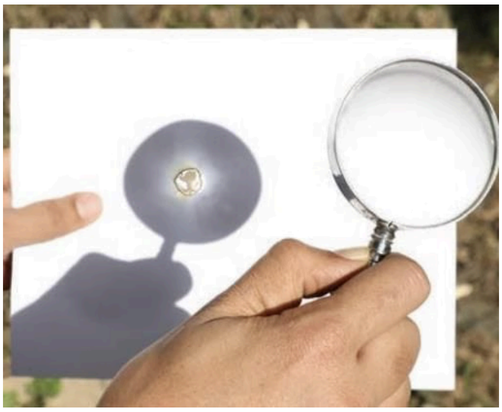


Candle (a) burning (b) covered with a glass tumbler

What else is needed to start combustion?

Besides a **fuel** and **oxygen**, **heat** is needed to start combustion. The minimum temperature at which a substance starts burning is called its **ignition temperature**.

A lighted match supplies the heat needed to raise paper to its ignition temperature and it burns.
Concentrated sunlight (using a magnifying glass) can also heat paper to its ignition temperature and set it on fire.



Paper catching fire

Fascinating fact

Have you seen tiny glowing insects in gardens at night? They are called **fireflies**. Their light is produced by a chemical reaction inside their bodies and is an example of **bioluminescence**. This glow is produced with very little heat.

Science and society - safety

What to do if someone's clothes catch fire?

Wrap the person in a **blanket or cloth** to cut off the air supply and stop the flames. This is safer than trying to blow out the fire. Use thick, non-synthetic material where possible.

Important: Do not use synthetic blankets or clothing because they may melt and stick to the skin, causing more severe injuries.

MULTIPLE CHOICE QUESTION

Try yourself: What substance is formed when iron rusts?

A Water

B Iron oxide

C Carbon dioxide

D Magnesium oxide

View Solution

Can Physical and Chemical Changes Occur in the Same Process?

1. Burning a candle: physical changes

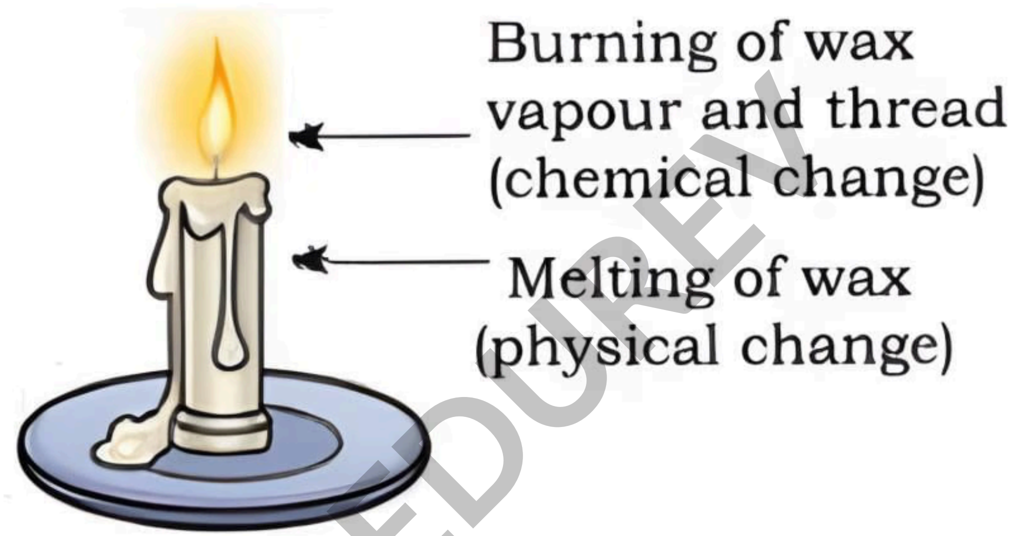
Melting of wax: Heat from the flame melts the solid wax to liquid - a physical change because the chemical identity of wax does not change.

Evaporation: Liquid wax near the wick vaporises and travels up the wick as vapour - a physical change.

Solidification: Wax that drips and cools solidifies back into a solid - another physical change.

2. Burning a candle: chemical change

Combustion of wax vapour: The vapour burns in oxygen to produce **carbon dioxide**, **water**, heat and light. New substances are formed, so this part is a chemical change.



Conclusion: Burning a candle involves both physical changes (melting, evaporation, solidification of wax) and chemical changes (oxidation of vapour forming new substances).

Know a scientist: Michael Faraday

Michael Faraday, a nineteenth-century scientist, used candles to explain physical and chemical processes in simple language. His popular lectures titled **"Chemical History of a Candle"** describe melting, vapourisation and combustion clearly for learners.

Are Changes Permanent?

Reversible changes

Definition: Reversible changes are those where the original substance or object can be restored by simple physical methods.

Examples:

- **Melting ice:** Ice melts to water; freezing the water reforms ice.
- **Boiling water:** Water becomes steam; condensing steam returns liquid water.
- **Folding paper:** Folding and then unfolding restores the original shape.

Irreversible changes

Definition: Irreversible changes are those where the original substance or object cannot be restored by simple physical means.

Examples:

Chopping vegetables: Cut pieces cannot be joined back to form the whole vegetable.

Making popcorn: Corn kernels transform into popcorn and cannot revert to kernels.

Burning wood: Wood turns to ash; ash cannot become wood again.

Are All Changes Desirable?

Desirable changes

Desirable changes are useful or beneficial to us.

Examples:

Milk to curd: Milk ferments to form curd which is edible and nutritious.

Ripening fruits: Fruits change colour, become sweeter and edible.

Cooking food: Raw ingredients change to cooked food which is easier to digest and safer to eat.

Cutting fruits: Makes them ready to eat.

Undesirable changes

Undesirable changes are harmful or unwanted.

Examples:

Rusting of iron: Damages tools and structures.

Food decay: Spoils stored food, making it unfit to eat.

Some changes may be useful in one situation and harmful in another. For example, decomposition of kitchen waste is undesirable when it causes foul smell and disease, but the same process is useful for making compost for plants.

Environmental impact

Human activities such as burning fossil fuels in vehicles and industries release **carbon dioxide** and other pollutants into the air.

These emissions increase atmospheric pollution and contribute to long-term changes like **climate change**.

MULTIPLE CHOICE QUESTION

Try yourself: What happens to wax when it is melted by a candle flame?

A It forms a new substance.

B It evaporates into vapor.

C It becomes liquid without a new substance.

D It turns into ash.

View Solution

Some Slow Natural Changes

Weathering of rocks

Weathering is the process by which rocks break down into smaller pieces (sediments) through physical and chemical processes. Over long periods, these sediments contribute to soil formation.

Types of weathering

- **Physical weathering:** Caused by temperature changes (expansion and contraction), plant roots growing into cracks, or water freezing and expanding in crevices. These actions break rocks into smaller pieces like sand and gravel.
- **Chemical weathering:** Occurs when water or dissolved chemicals react with minerals in rocks and change their composition. For example, iron-containing rocks may form a reddish layer of iron oxide (similar to rust) when exposed to water and air.

Both types of weathering over long periods help form soil, which is essential for plant growth and life on Earth.

Erosion

Erosion is the movement or removal of weathered rock, soil and sediments by natural forces such as flowing water, wind or glaciers.

Examples and effects

- **Sand on riverbeds:** Large rocks and stones are worn down and moved by rivers, producing fine sand and pebbles.
- **Smoothing of river rocks:** Flowing water rubs stones against each other, making them smooth over time.
- **Landslides:** Rapid erosion during landslides moves large amounts of soil and rock downhill; this is a fast physical change.
- When wind or water slows down, the carried sediments settle; over thousands of years they can be compacted and hardened into new rocks.

These changes are usually slow and often irreversible on human timescales.

MULTIPLE CHOICE QUESTION

Try yourself: What is weathering?

A Creating new rocks from sediments.

B A type of chemical reaction.

C Breaking down rocks into smaller pieces.

D The process of moving sediments.

View Solution

Terms to Remember

Physical changes: Only shape, size or state change; no new substances are formed. Examples: melting ice, boiling water, folding clothes.

Chemical changes: New substances are formed. Examples: rusting, burning, lime water turning milky on passing carbon dioxide.

Rusting: Iron reacts with air and water to form rust (iron oxide), a chemical change.

Combustion: Burning in oxygen that produces heat and light; requires fuel, oxygen and sufficient heat (ignition temperature).

Fireflies: Glow due to a chemical reaction called **bioluminescence**, which produces light with very little heat.

Burning a candle: Involves both physical changes (wax melting and evaporating) and a chemical change (vapour burning to form carbon dioxide and water).

Reversible changes: Changes that can be reversed by simple means (e.g., melting and freezing of water).

Irreversible changes: Changes that cannot be easily reversed (e.g., cutting, burning).

Desirable changes: Useful or beneficial (e.g., cooking, making curd, ripening of fruits).

Undesirable changes: Harmful or unwanted (e.g., rusting, food spoilage).

Weathering: Breakdown of rocks into soil by physical and chemical processes.

Erosion: Movement of soil and sediments by wind, water or ice, shaping the land.

Michael Faraday: A scientist who explained physical and chemical processes using everyday examples such as candles in his lectures **"Chemical History of a Candle"**.

You can practice questions from this chapter here: **HOTS Questions: Changes Around Us: Physical and Chemical** (https://edurev.in/studytube/Class-7-Science-Chapter-5-HOTS-Questions--Changes-Around-Us-Physical-and-Chemical/87c22e94-f79c-4b9d-a748-63a5806c5e26_t)